USER MANUAL

AquaSYNC

Smart IoT Water Quality & Level Management System

Aqua Sync

Sense the flow. Save the future.

Designed & Developed by:

[TEAM AQUA SYNC]

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1. Introduction

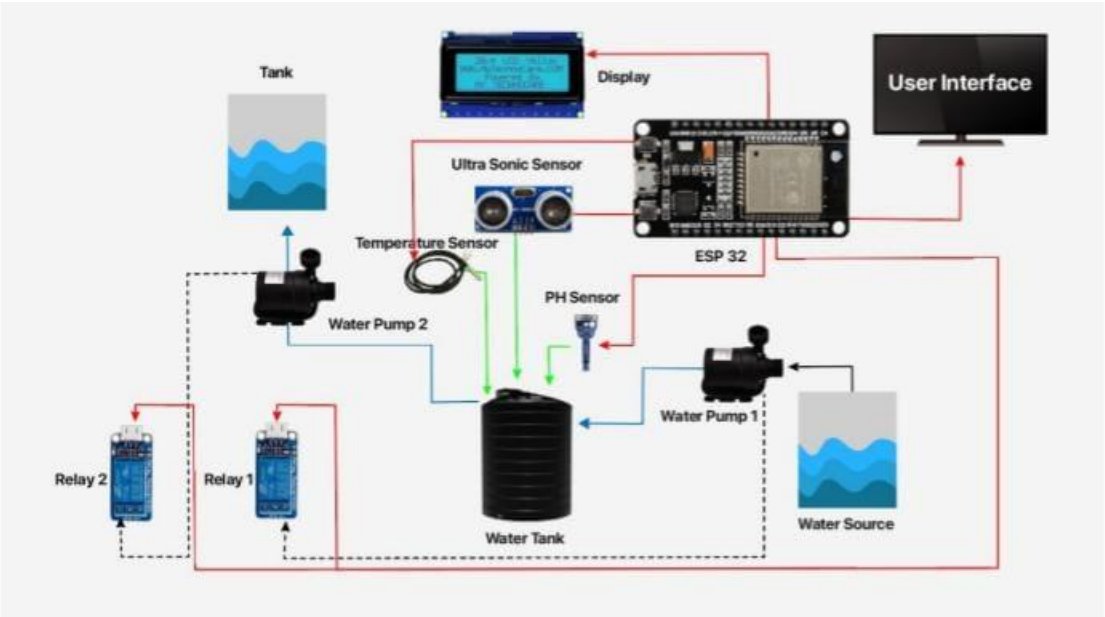
Welcome to AquaSYNC — an advanced, integrated solution for automated water tank management. Combining precision hardware sensing with cutting-edge Internet of Things (IoT) technology, AquaSYNC ensures your water supply is continuously monitored, managed, and maintained at optimal levels and quality.

This system monitors two separate water tanks, tracks vital parameters such as Temperature and pH, and automates water pumping and pH correction. With the integration of the Blynk 2.0 platform, you retain complete oversight and control from anywhere in the world via your smartphone.

2. Key Features

- **Dual-Tank Ultrasonic Monitoring:**
  - Real-time level measurement of both the Main storage tank and the Source tank.
- **Water Quality Sensing:**
  - Precision monitoring of water pH level and Temperature.
- **Intelligent Automation:**
  - Automated refilling of the Main tank when levels drop below 20%.
  - Automated pump stop at 80% full to prevent overflow.
  - Automated pH dosing regulation (activated when pH is outside the 6–8 range).
- **Sensor Noise Suppression (Anti-Interference):**
  - Advanced logical locking prevents erratic pH and Temperature readings caused by electrical interference while the heavy-duty water pump is operational.
- **Hardware Safety Interlocks:**
  - The pH pump is logically disabled whenever the high-voltage water pump is running to prevent dosing errors.
- **IoT Integration (Blynk 2.0):**
  - Remote real-time monitoring and control through a custom mobile dashboard.
- **Multi-Mode Operation:**
  - Switch seamlessly between Automatic and Manual control modes for both pumps.
- **Hardware Override:**
  - Physical toggle switches provide immediate manual control over pumps, bypassing software logic in emergencies.
- **Visual & Audio Alarms:**
  - On-board LEDs and Buzzers provide instant alerts for critical conditions (Source Low, Main Low, High Temp).

3. System Hardware Overview



3.1

Hardware Components Guide

Referencing the numbered labels in Figure 2 above:

1. Main Controller (Arduino Uno) — Master controller responsible for sensor data processing and LCD output.
2. IoT Gateway (ESP32) — Wi-Fi enabled controller managing system logic, Blynk communication, and relay activation.
3. LCD Display (20x4) — Provides a comprehensive physical interface for real-time data visualization without requiring a smartphone.
4. Ultrasonic Sensors (HC-SR04) — Non-contact sensors that measure water levels in both the Main and Source tanks.
5. pH Sensor (SEN0161) — Measures the acidity or alkalinity of the water (0–14 scale).
6. Temperature Sensor (DS18B20) — Waterproof probe for accurate fluid temperature measurement.
7. Relay Modules — Opto-isolated switches controlling the 230V Water Pump (P1) and the low-voltage Dosing Pump (P2).
8. Status LEDs — Green (System OK), Red (Warning/Alarm), Yellow (Temperature Alert).
9. Hardware Override Switches — Industrial-grade toggle switches for direct pump control, bypassing all software logic.

4. On-Board LCD Display Interface

The AquaSYNC unit features a high-definition 20x4 LCD display that provides immediate system feedback — no smartphone required.

4.1 LCD Screen Layout & Data Points

Row	Display Example	Description
1	Main: 65% Src: 90%	Main & Source Level: Current water levels as percentage (0–100%).
2	pH: 7.2 T: 28.5C	Water Quality: Live pH (0.0–14.0) and Temperature (0.0–80.0°C).
3	P1: OFF P2: ON	Pump Status: Whether Water Pump (P1) or pH Pump (P2) are active.
4	System Active	Status Bar: General system status or current alarm messages.

5. System Alarms & Visual Indicators

AquaSYNC uses a combination of LED indicators and an on-board buzzer to alert you to critical system states without requiring app access.

Condition	Visual Indicator	Audio Alert	Description
System OK	Green LED ON	No Sound	All parameters within normal ranges.
Source Low (<10%)	Red LED Flashing	Fast Beeping	Source water insufficient. Pump P1 auto-disabled.
Main Very Low (<5%)	Red LED Flashing	Slow Beeping	Critical low level in storage. Immediate action required.
High Temp (>30°C)	Yellow LED ON	Intermittent Beep	Water temperature exceeded safe limits.

6. Blynk IoT Mobile Dashboard

For showcase purposes, the Blynk 2.0 mobile app is the primary remote monitoring tool. Ensure the ESP32 is connected to the designated Wi-Fi network before use.

6.1 Dashboard Elements Explained

Referencing the layout labels in Figure 4 above:

- **A — Main Tank Level (Gauge):**
  - Visual gauge showing current storage tank capacity as a percentage.
- **B — Source Tank Level (Bar):**
  - Vertical level bar showing available source water (%).
- **C — Temperature (Gauge):**
  - Current water temperature displayed in degrees Celsius.
- **D — pH Level (Gauge):**
  - Real-time pH measurement on the 0–14 scale.
- **E — Pump Status LEDs:**
  - Virtual green and blue indicator lights that illuminate when their respective physical pumps are active.
- **F — Mode Switches:**
  - Toggle between AUTO (logic-controlled) or MANUAL modes for Pump 1 (P1) and Pump 2 (P2) independently.
- **G — Power Switches:**
  - In MANUAL mode only, these provide direct ON/OFF control of the pumps from the app.
- **Push Notifications:**
  - Critical alerts (e.g., "Source Tank water is below 10%!") appear as push notifications directly on your smartphone screen.

7. Operational Modes Explained

A. Water Pump (P1) — Control Logic

Mode	Condition	Result
AUTO	Main Tank < 20% AND Source > 10%	Pump starts (ON) to refill storage.
AUTO	Main Tank > 80% OR Source < 10%	Pump stops (OFF).
MANUAL	App Power Switch activated	Direct control via App, regardless of sensors.
OVERRIDE	Physical switch on box set to ON	Hardware activation. Bypasses all software. Use with caution.

⚠ **IMPORTANT:** pH and Temperature sensors are 'frozen' to their last known good state while P1 is running. This is by design — it prevents false readings caused by electrical interference from the heavy-duty pump motor. Values will resume updating automatically once P1 stops.

B. pH Dosing Pump (P2) — Control Logic

Mode	Condition	Result
ANY	Water Pump (P1) is running	P2 IS AUTOMATICALLY DISABLED — Safety Lock.
AUTO	P1 is OFF AND (pH < 6 OR pH > 8)	Dosing pump starts (ON).
AUTO	pH is between 6 and 8	Pump stops (OFF).
MANUAL	P1 is OFF AND App Switch activated	Direct control via App.

8. Basic Troubleshooting

If you encounter unexpected behavior, consult the table below before contacting support.

Problem	Potential Cause	Solution
LCD Screen is Blank	Power disconnect.	Check 12V DC input and Uno connection.
pH/Temp Readings 'Stuck'	Water pump (P1) is running.	This is NORMAL — logic prevents electrical noise. Values update when P1 stops.
ESP32 'Connecting...'	Incorrect Wi-Fi SSID/Password. Router off.	Verify credentials (Galaxy_S25_FE, Tharu2004). Ensure router/hotspot is ON.
Relay does not click	5V PSU to Relay disconnected. ESP32 pin loose.	Verify external 5V power supply. Check wiring to Pins 27/14.